Telecom Customer Churn Prediction Using Logistic Regression

This project focuses on predicting customer churn in a telecom company using Machine Learning. The objective is to identify customers who are likely to leave the service based on account details, call usage, service plans, and customer service interactions.

Dataset Features
• Account_Length
• Vmail_Message
• Day_Mins, Eve_Mins, Night_Mins, Intl_Mins
• CustServ_Calls
• Intl_Plan
• Vmail_Plan
• Churn (Target Variable)

Technologies Used
• Python
• Pandas
• Matplotlib
• Scikit-learn

Machine Learning Algorithm
Logistic Regression is a supervised machine learning algorithm used to classify customers as churn or non-churn based on telecom customer data.

Project Workflow
1. Import Libraries
2. Load Dataset
3. Data Preprocessing
4. Convert Text Columns to Numerical Values
5. Generate Churn Distribution Graph
6. Split Dataset into Training and Testing Sets
7. Train Logistic Regression Model
8. Predict Customer Churn
9. Evaluate Model Accuracy

Conclusion
The Telecom Customer Churn Prediction project successfully uses Logistic Regression to identify customers likely to leave the telecom service. The model can help telecom companies improve customer retention strategies and customer satisfaction.